

## SIMATS ENGINEERING

### ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

#### COURSE OUTCOME COVERED

**CO2:** Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Scenario-Based Questions

#### Code of Conduct:

*I K Sreelatha(192425248)\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student: Sree K**

#### RUBRICS FOR CALCULATION:

| Criteria                     | Weightage | Level 4 (Exemplary)                                                       | Level 3 (Proficient)                               | Level 2 (Developing)                              | Level 1 (Beginning)                               | Marks (100) |
|------------------------------|-----------|---------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------|---------------------------------------------------|-------------|
| Understanding of Scenario    | 20%       | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps           | Partial understanding; misses key aspects         | Misinterprets or fails to understand the scenario |             |
| Identification of Key Issues | 20%       | Accurately identifies all relevant issues and factors involved            | Identifies most key issues with minor omissions    | Identifies limited issues; some irrelevant points | Unable to identify key issues                     |             |
| Application of Concepts      | 25%       | Applies appropriate concepts effectively to address the scenario          | Applies concepts with minor errors                 | Limited or partially correct application          | Unable to apply relevant concepts                 |             |
| Decision Making              | 20%       | Makes well-reasoned, logical decisions with strong rationale              | Makes reasonable decisions with some justification | Makes weak decisions with limited justification   | No clear decisions made or rationale provided     |             |
| Clarity of Response          | 15%       | Response is clear, structured, and logically presented                    | Generally clear with minor issues                  | Somewhat unclear or poorly structured             | Disorganized and difficult to understand          |             |

## Scenario-Based Questions

### 1. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

- (a) Interpret the scenario and explain the importance of boundary detection.
- (b) Identify key challenges in detecting boundaries.
- (c) Apply a suitable edge detection method.
- (d) Justify your selection with logical reasoning.
- (e) Present your explanation clearly and systematically.

### 2. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

- (a) Interpret the scenario and explain segmentation requirements.
- (b) Identify key features enabling separation.
- (c) Apply a suitable segmentation technique.
- (d) Justify your decision logically.
- (e) Present your response clearly.

### 3. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

- (a) Interpret the scenario and explain segmentation challenges.
- (b) Identify key issues such as intensity overlap.
- (c) Apply an appropriate advanced segmentation method.
- (d) Justify how the method handles complexity.
- (e) Provide a structured and clear explanation.

### 4. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

- (a) Interpret the scenario and explain why edges are broken.
- (b) Identify key issues affecting boundary continuity.
- (c) Apply a suitable method to improve edge connectivity.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

### 5. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

- (a) Interpret the scenario and define the combined requirement.
- (b) Identify key issues involving noise and edge preservation.
- (c) Apply an appropriate sequence of techniques.
- (d) Justify the order and selection with strong reasoning.
- (e) Present your response in a clear and structured format.

## 1. Object Boundary Detection

### (a) Interpret the scenario and explain the importance of boundary detection

Medical images such as **MRI (Magnetic Resonance Imaging)**, **CT (Computed Tomography)**, **Ultrasound**, and **X-ray** contain different tissues and organs. To analyze these images accurately, the boundaries between tissues must be identified. Boundary detection is the process of locating points where the image intensity changes sharply. These points usually represent the borders of organs, tumors, bones, or blood vessels.

Boundary detection is important because it helps doctors:

- Detect tumors and abnormal growths.
- Measure the size and shape of organs.
- Plan surgeries and radiation therapy.
- Track disease progression over time.
- Improve the accuracy of computer-aided diagnosis (CAD).

### (b) Identify key challenges in detecting boundaries

Medical images present several challenges:

- **Image Noise:** Noise introduced by imaging devices may create false edges.
- **Low Contrast:** Neighboring tissues often have very similar intensity values.
- **Blurred Boundaries:** Motion of the patient or imaging limitations can make edges unclear.
- **Complex Structures:** Organs have irregular shapes and overlapping structures.
- **Weak Edges:** Small tumors or thin blood vessels produce weak intensity changes.

### (c) Apply a suitable edge detection method

#### Canny Edge Detection

##### Algorithm Steps

1. Apply **Gaussian Filter** to remove noise.
2. Calculate image gradients using Sobel operators.
3. Perform **Non-Maximum Suppression** to obtain thin edges.
4. Apply **Double Thresholding** to classify strong and weak edges.
5. Use **Edge Tracking by Hysteresis** to connect valid edge segments.

### (d) Justify your selection

Canny Edge Detection is preferred because:

- High detection accuracy.
- Excellent noise suppression.
- Produces thin and continuous edges.
- Detects weak edges without creating many false edges.
- Widely used in medical image processing applications.

### **Advantages**

- High accuracy
- Good localization
- Resistant to noise
- Continuous boundaries

### **Disadvantages**

- Slightly slower than Sobel or Prewitt.
- Requires threshold selection.

### **Applications**

- Tumor detection
- Brain MRI analysis
- Organ segmentation
- Blood vessel extraction
- Bone fracture detection

## 2. Simple Segmentation

### (a) Interpret the scenario

Segmentation divides an image into meaningful regions. In this scenario, the object is clearly distinguishable from the background because their intensity values differ significantly.

Example:

- White coin on a black table.
- Bright cell on a dark microscope image.

### (b) Identify key features

Important characteristics include:

- High contrast between object and background.
- Uniform intensity inside the object.
- Uniform background.
- Minimal noise.

### (c) Apply a suitable segmentation technique

#### Global Thresholding

The threshold value  $T$  separates foreground and background.

Mathematically,

If

**Pixel Intensity  $> T \rightarrow$  Object**

Else

**Pixel Intensity  $\leq T \rightarrow$  Background**

Result:

A binary image where

- White = Object
- Black = Background

### (d) Justification

Thresholding is selected because:

- Easy to implement.
- Very fast.

- Requires little computation.
- Suitable for real-time image processing.
- Works well when illumination is uniform.

### **Advantages**

- Simple algorithm
- Fast execution
- Low memory usage
- Good for high-contrast images

### **Limitations**

- Poor performance under uneven lighting.
- Cannot handle overlapping intensities.
- Sensitive to noise.

### **Applications**

- OCR (Optical Character Recognition)
- Industrial inspection
- Cell counting
- Barcode detection
- Document image processing

### **3. Complex Segmentation**

#### **(a) Interpret the scenario**

In many real-world images, different objects may have almost identical intensity values. As a result, simple thresholding fails because pixels belonging to different objects look similar.

Examples include:

- Brain MRI with gray matter and white matter.
- Satellite images.
- Industrial components with similar colors.

#### **(b) Identify key issues**

Major challenges are:

- Overlapping intensity values.
- Touching or connected objects.
- Uneven illumination.
- Image noise.
- Irregular object shapes.

#### **(c) Apply an advanced segmentation technique**

##### **Marker-Controlled Watershed Segmentation**

Procedure:

1. Smooth the image.
2. Compute gradient magnitude.
3. Identify foreground markers.
4. Identify background markers.
5. Apply Watershed transformation.
6. Generate separated regions.

#### **(d) Justification**

Watershed is effective because:

- Separates touching objects.
- Produces closed boundaries.
- Marker control reduces over-segmentation.

- Handles irregular shapes efficiently.

### **Advantages**

- High segmentation accuracy.
- Works for connected objects.
- Produces complete object boundaries.

### **Limitations**

- Can over-segment without markers.
- Computationally expensive.

### **Applications**

- Medical imaging
- Cell segmentation
- Satellite image analysis
- Object recognition
- Material inspection



## **4. Broken Edges**

### **(a) Interpret the scenario**

Sometimes edge detection algorithms produce incomplete boundaries. Instead of continuous outlines, objects appear as disconnected edge fragments.

This occurs because:

- Weak edge intensity.
- Noise.
- Poor illumination.
- Threshold settings.
- Blurred images.

### **(b) Key issues**

- Missing edge pixels.
- Boundary gaps.
- Noise interference.
- Weak gradients.
- Incomplete contours.

### **(c) Apply a suitable method**

#### **Morphological Closing**

Closing consists of:

**Closing = Dilation + Erosion**

#### **Dilation**

- Expands object boundaries.
- Connects nearby edge fragments.

#### **Erosion**

- Restores object size.
- Removes unnecessary thickening.

### **(d) Justification**

Closing is ideal because:

- Connects broken edges.
- Fills small holes.
- Preserves object shape.

- Improves contour continuity.

### **Advantages**

- Connects discontinuous edges.
- Removes tiny holes.
- Improves segmentation quality.

### **Applications**

- Character recognition
- Fingerprint analysis
- Shape detection
- Medical image analysis

## **5. Combined Requirement**

### **(a) Interpret the scenario**

Industrial inspection systems inspect products for manufacturing defects such as:

- Cracks
- Scratches
- Holes
- Surface damage

Images captured by cameras often contain noise. The system must remove this noise while preserving important edges so defects remain visible.

### **(b) Identify key issues**

Challenges include:

- Salt-and-pepper noise.
- Gaussian noise.
- Fine defect edges.
- False edge detection.
- Edge blurring due to filtering.

### **(c) Apply an appropriate sequence**

#### **Step 1: Median Filter**

Purpose:

- Removes salt-and-pepper noise.
- Preserves edges better than averaging filters.

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#### **Step 2: Canny Edge Detection**

Purpose:

- Detects accurate and continuous edges.
- Removes weak false edges.

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#### **Step 3: Morphological Closing**

Purpose:

- Connects broken edges.
- Produces complete defect boundaries.

#### **(d) Justification**

The sequence is important because:

##### **Median Filter First**

- Noise is removed before edge detection.
- Prevents false edge formation.

##### **Canny Second**

- Detects precise object boundaries.
- Uses the cleaned image for better accuracy.

##### **Morphological Closing Last**

- Repairs discontinuous boundaries.
- Produces complete defect outlines.

This order ensures maximum accuracy and reliable defect detection.

##### **Advantages**

- High inspection accuracy.
- Better defect localization.
- Noise-free edge detection.
- Continuous object boundaries.
- Suitable for automated quality control.

##### **Applications**

- Printed Circuit Board (PCB) inspection
- Metal surface crack detection
- Textile defect inspection
- Bottle and container quality inspection
- Automobile component inspection
- Semiconductor wafer inspection

## Overall Comparison

| Problem                           | Technique                                       | Why Used                                          | Main Advantage                       |
|-----------------------------------|-------------------------------------------------|---------------------------------------------------|--------------------------------------|
| Object Boundary Detection         | Canny Edge Detection                            | Detect tissue boundaries                          | Accurate, thin, continuous edges     |
| Simple Segmentation               | Global Thresholding                             | Separate object from background                   | Fast and simple                      |
| Complex Segmentation              | Marker-Controlled Watershed                     | Separate overlapping regions                      | Handles touching objects effectively |
| Broken Edges                      | Morphological Closing                           | Connect discontinuous edges                       | Produces continuous boundaries       |
| Noise Removal + Edge Preservation | + Median Filter → Canny → Morphological Closing | Remove noise while preserving and repairing edges | Reliable defect detection            |